

Business Case

Project: Financial and Operational Analysis of U.S. Hospitals Using Provider Cost Report Data (2011–2022)

1. Project Purpose & Business Need

This project aims to analyze **U.S. hospital financial and operational performance** using official **CMS Hospital Provider Cost Report data** from 2011–2022.

Hospitals operate in a highly pressured environment with rising costs, changing reimbursement models, and regulatory requirements. Decision-makers often lack a **consolidated, analytics-ready view** of key metrics such as:

- Total costs and total charges
- Cost-to-charge ratio
- Beds, FTE employees, interns/residents
- Patient days and discharges
- Medicare/Medicaid payments

Currently, this information is spread across multiple annual files and complex CMS worksheets. This creates:

- Time-consuming manual analysis
- Inconsistent reporting across years
- Difficulty comparing hospitals across states, regions, and urban/rural categories

Business Need:

The organization requires a **clean, integrated dataset and analytical framework** to support:

- Performance benchmarking across hospitals
- Identification of efficiency gaps and cost drivers
- Evidence-based financial and operational planning

2. Business Case Summary

The project will deliver an **analytics-ready dataset** and a set of **visual and statistical analyses** that enable deep insights into hospital performance over twelve years.

Key components:

- Integrate 12 years of CMS cost reports into a **single, standardized dataset** (2011–2022, 73,974 records, 118 features).
- Assess data quality, handle missing values, outliers, and data type issues.
- Build dashboards and models to explain and predict financial and operational performance.

This supports **better financial planning, risk assessment, and policy analysis** for stakeholders working with hospital systems or health authorities.

3. Strategic Alignment

This project aligns with organizational and academic objectives by:

- Supporting **data-driven decision-making** in healthcare finance and operations.
- Leveraging **public, government-verified data** for transparent analytics.
- Enabling **trend analysis** over more than a decade to understand sustainability and efficiency patterns.
- Providing a solid foundation for **advanced analytics** (regression, clustering, forecasting) in line with analytics program learning outcomes.

4. Expected Benefits

4.1 Tangible Benefits

- **Time savings:**
 - Reduced manual effort integrating yearly CMS files into one consistent structure.
- **Improved analysis quality:**
 - Standardized data types, cleaned missing values, and treated outliers improve reliability of dashboards and models.
- **Reusable dataset:**
 - A unified dataset can be reused for future projects, courses, or organizational analyses without repeating the integration work.

4.2 Intangible Benefits

- **Better insight into hospital efficiency and sustainability** (e.g., how beds, FTEs, and patient days relate to total costs and charges).
- **Enhanced decision support** for stakeholders interested in rural vs. urban hospitals, state-level differences, or payer dependence (Medicare/Medicaid).
- **Improved transparency** by using official CMS data sources.
- **Learning and capability building** for the analytics team in data engineering, visualization, and modeling.

5. High-Level Scope

5.1 In Scope

- Use the **combined_hospital_cost_reports_2011_2022.csv** file as the primary dataset (73,974 rows, 118 columns).
- Perform a **full data assessment** (duplicates, schema consistency, missing values, outliers, negative values, ZIP code integrity).
- Standardize key variables:
 - Identification: report ID, provider ID (CCN)
 - Demographics: state, county, ZIP, urban/rural indicator, CBSA

- Operational: beds, FTE employees, interns/residents, patient days, discharges
- Financial: total charges, total costs, revenues, payment types, cost-to-charge ratio
- Temporal: fiscal year begin/end
- Conduct data cleaning and pre-processing:
 - Drop or impute highly missing variables
 - Treat outliers (e.g., winsorization /log transforms for costs and charges)
 - Convert ZIP codes to strings, ensure proper date types
 - Engineer ratios (e.g., cost/charge, Medicare dependence %).
- Develop **exploratory visualizations and dashboards** (e.g., trends by year, state, urban/rural, hospital size).
- Perform **basic modeling**:
 - Regression (e.g., drivers of total costs or cost-to-charge ratio)
 - Clustering (e.g., hospital segments by financial/operational patterns).

5.2 Out of Scope

- Patient-level analysis (no patient identifiers in this data).
- System changes or operational interventions at real hospitals.
- Real-time or streaming data integration.
- Extensive external data (e.g., adding new datasets beyond CMS cost reports) unless approved as a later phase.

6. Key Deliverables

1. **Cleaned, integrated CMS hospital dataset (2011–2022)**
2. **Data Assessment Report** (already drafted, refined as needed) summarizing:
 - Completeness, consistency, and reliability
 - Missing values, outliers, and data integrity diagnostics
3. **Data preparation and feature engineering notebook(s)** (Python).
4. **Visualization layer**:
 - Dashboards/plots showing trends, distributions, and benchmarks.
5. **Analytical outputs**:
 - Regression results describing drivers of financial performance
 - Clustering or segmentation of hospitals (e.g., by size, performance)
6. **Final report or presentation**:
 - Executive summary of insights
 - Recommendations for decision-makers or policymakers

7. High-Level Risks & Assumptions

7.1 Risks

- **Missing Values:**
 - Some specialized variables (e.g., psychiatric, teaching metrics) have 85–95% missingness, which may limit their use in models.
- **Outliers and Extreme Values:**
 - Costs and charges are highly skewed with multi-million outliers; inappropriate handling could distort models.
- **Negative Values:**
 - Negative values (e.g., refunds/adjustments) can complicate ratio calculations and must be interpreted carefully.
- **Schema / ID Changes Over Time:**
 - Hospital mergers or ID changes may affect longitudinal tracking across years.

7.2 Assumptions

- CMS data definitions remain consistent enough to support cross-year integration.
- All work is done at **facility-level**, with no patient-level confidential data.
- Tools such as Python, Tableau/Power BI, and required computing resources are available.
- Team members will contribute to cleaning, EDA, and modeling according to plan.

8. Ethical & Compliance Considerations

- Data is **public, facility-level CMS data** with no patient identifiers, minimizing privacy risk.
- Analyses should avoid unfairly labeling facilities as "good" or "bad" without context (e.g., rural constraints, teaching roles).
- Any findings should be framed as **support for improvement** rather than punitive judgment.
- Documentation of assumptions, limitations, and potential bias (e.g., rural vs. urban representation) will be included.

9. Financial Justification (Conceptual)

While this is an academic project, if implemented in a real organization, the financial justification would be based on:

- **Cost:**
 - Analyst time for data integration, cleaning, and dashboard/model development.
- **Benefits:**
 - Time saved in recurring reporting across multiple hospitals and years.
 - Better decisions around staffing, capacity, and cost control.
 - Earlier identification of financially distressed hospitals.

This could be evaluated using the **Benefit–Cost Ratio, NPV, and Payback Period** concepts discussed in Week 3 (e.g., if analytics tools save X analyst hours/year at rate Y, compared to initial project effort).

Week 3 -2

10. Recommendation

Given:

- The **high reliability and richness** of CMS Hospital Provider Cost Report data,
- The **strong analytical readiness** after moderate cleaning,
- The clear alignment with **strategic and educational goals in healthcare analytics**, and
- The potential for **significant insight into cost efficiency and operational performance**

11. Project Sponsor & Stakeholders

Project Sponsor

- **Professor Manjari Maheshwari** – Capstone Supervisor, St. Clair College
Provides oversight, approves project direction, ensures academic alignment, and validates deliverables.

Primary Stakeholders

Role	Name	Responsibilities
Team Leader & Data Integration Lead	<i>Manjunath Muthineni</i>	Oversees project execution, ensures timely progress, leads dataset integration, coordinates team responsibilities, manages GitHub structure.
Finance Subject Matter Expert (SME)	<i>Neha</i>	Leads financial metric interpretation, contributes to financial dashboard design and validation.
Data Cleaning Lead	<i>Abhishekh Choudhary</i>	Responsible for dataset cleaning, missing value treatment, outlier handling, and preparing the analysis-ready data.
Visualization & Analytics Lead	<i>Krishna Chaitanya Venuturumilli</i>	Develops EDA, operational dashboard, Power BI structure, readme/requirements updates, and analytical insights.
Documentation Lead	<i>Rajesh Thota</i>	Prepares Business Case, Meeting Minutes, Project Charter, BRD, and final report formatting.

Secondary Stakeholders

- **St. Clair College – Data Analytics Program Faculty**
Ensures project meets academic and program learning outcomes.
- **Healthcare Analytics Community/Researchers**
Potential users of insights, dashboards, and reusable dataset structure.
- **Future Students & Stakeholders**
May reuse the integrated dataset for future academic projects or research.